VIT BHOPAL UNIVERSITY

School of Computer Science & Engineering

CENTRAL LIBRARY MANAGEMENT SYSTEM

A Desktop Application using Python, CustomTkinter & SQLite

A Final Project Report submitted in partial fulfilment
of the requirements for the degree of

Bachelor of Technology
in
Artificial Intelligence & Machine Learning

Submitted by:	Anshuman Joshi
Registration No:	26BAI10158
Programme:	B.Tech – AI & ML
Guide:	Dr. Gunjan Ansari
Academic Year:	2026-2027

CERTIFICATE

This is to certify that **Anshuman Joshi**, Registration No. **26BAI10158**, a student of the Bachelor of Technology in Artificial Intelligence & Machine Learning at VIT Bhopal University, has successfully completed the project titled **"Central Library Management System"** under the supervision of **Dr. Gunjan Ansari** during the academic year 2026–27.

The work embodied in this report is original and has been carried out by the candidate.

_____

Dr. Gunjan Ansari
Project Guide

Dr. Rajit Nair
Head of Department

DECLARATION

I hereby declare that the project report titled **"Central Library Management System"** submitted to VIT Bhopal University is my own original work.

Anshuman Joshi
Reg. No: 26BAI10158
B.Tech AI & ML
Date: 30 September 2026

ABSTRACT

The Central Library Management System is a desktop application developed for VIT Bhopal University to automate library operations. It replaces the manual register-based book issuing process with a digital workflow covering cataloguing, circulation, reservations, fine calculation and report generation.

Built with **Python 3**, **CustomTkinter** and **SQLite3**, the system provides three role-based portals — Student, Faculty and Librarian. Students borrow up to 2 books for 14 days, faculty up to 5 books for 60 days, and librarians have full administrative control.

Fines are calculated automatically at **Rs 2 per day** for late returns, and borrowing is blocked when unpaid fines exceed **Rs 50**. A reservation queue automatically promotes the next member when a copy is returned. Librarians can export multi-sheet Excel workbooks and CSV reports.

TABLE OF CONTENTS

1. INTRODUCTION **7**

2. TECHNOLOGY REVIEW **8**

3. SYSTEM ANALYSIS **8**

4. SYSTEM DESIGN **9**

4.1 Architecture Diagram 9

4.2 ER Diagram 11

4.3 Class Diagram 12

4.4 Use Case Diagram 13

4.5 DFD Level 0 14

4.6 DFD Level 1 15

4.7 DFD Level 2 16

4.8 Sequence Diagrams 17

4.9 Activity Diagram 19

4.10 Component Diagram 20

4.11 Deployment Diagram 21

4.12 State Diagrams 22

4.13 Package Diagram 26

5. DATABASE DESIGN **27**

6. IMPLEMENTATION **27**

7. TESTING **39**

8. RESULTS AND CONCLUSION **40**

1. INTRODUCTION

1.1 Problem Statement

The VIT Bhopal Central Library currently relies on manual record-keeping for issuing and returning books. This leads to lost books, incorrect fines, no remote availability check, and slow reports. A digital system is needed to automate these operations.

1.2 Objectives

- Provide a single platform for books, users, issues and returns.
- Enforce borrowing limits and due dates automatically.
- Calculate fines at Rs 2 per day after the due date.
- Allow students and faculty to search and reserve books online.
- Generate statistical reports in Excel and CSV.
- Provide separate role-based portals.

1.3 Scope

Cataloguing, circulation, reservations, fines, notices, purchase requests and reporting. Standalone desktop app with local SQLite storage. No online payments, SMS, or university-system integration.

1.4 Target Users

Role	Borrowing Limit	Loan Period	Special Permissions
Student	2 books	14 days	Search, reserve, view notices
Faculty	5 books	60 days	Student features + purchase requests
Librarian	—	—	Full administrative access

2. TECHNOLOGY REVIEW

2.1 Python 3

High-level, interpreted language used for the complete application logic (Python 3.10+).

2.2 CustomTkinter

Modern wrapper around Tkinter providing rounded corners, dark/light themes and modern widgets for the whole UI.

2.3 SQLite3

Lightweight file-based relational database storing everything in a single file (`library.db`). Built into Python standard library.

2.4 Matplotlib

Used for bar charts on the librarian dashboard (issues over 14 days, most-borrowed sections).

2.5 openpyxl

Used to export multi-sheet styled Excel workbooks.

3. SYSTEM ANALYSIS

3.1 Existing System

Manual paper-register system. Fines calculated by hand, no remote availability, reports take hours.

3.2 Proposed System

Desktop app with three role-based portals, automatic fines, real-time search, reservation queue, oneclick report export.

3.3 Functional Requirements

- Registration and login with role selection
- Catalogue search by title, author, ISBN, publisher
- Issue / return with automatic due dates
- Fine calculation at Rs 2 per day
- Reservation queue with auto-promotion
- Notice board and purchase-request workflow
- Report export as XLSX and CSV

3.4 Non-Functional Requirements Sub-

- second response time
- Data integrity via foreign keys
- Dark / Light theme support
- Portable — Python 3.10+ only

3.5 Feasibility Study

- **Technical:** All technologies free and open source
- **Economic:** No licensing cost
- **Operational:** Simple UI, minimal training

4. SYSTEM DESIGN

4.1 Architecture Diagram

Three-layer architecture — Presentation (CustomTkinter), Business Logic (database.py), Data (SQLite).

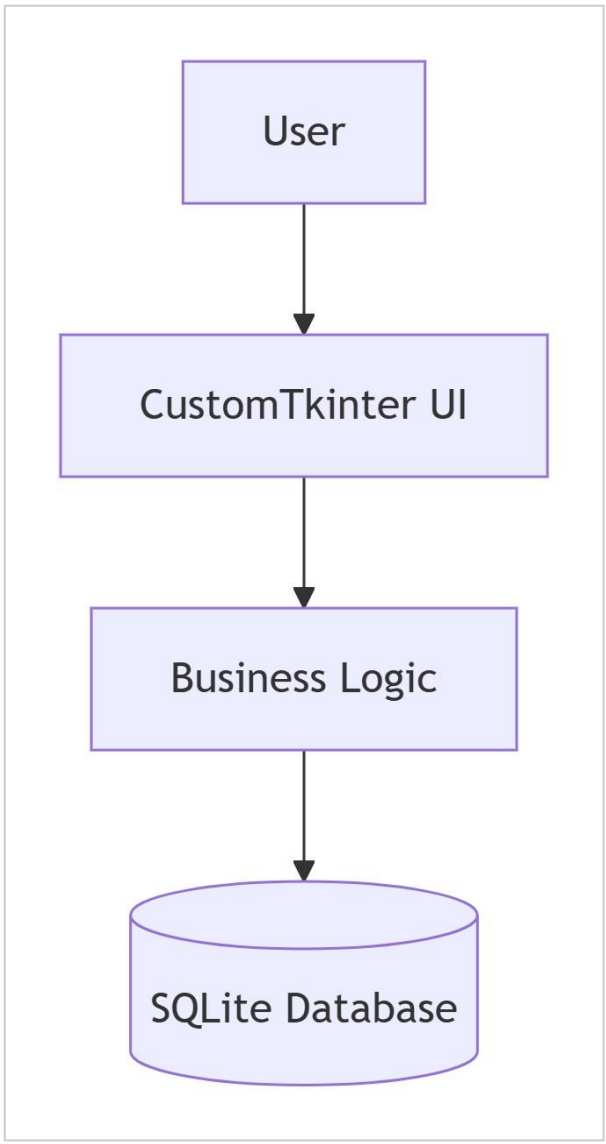


Figure 1: Three-layer System Architecture

4.2 ER Diagram

Six tables — USERS, BOOKS, ISSUES, RESERVATIONS, NOTICES and REQUESTS. Foreign keys enforce referential integrity.

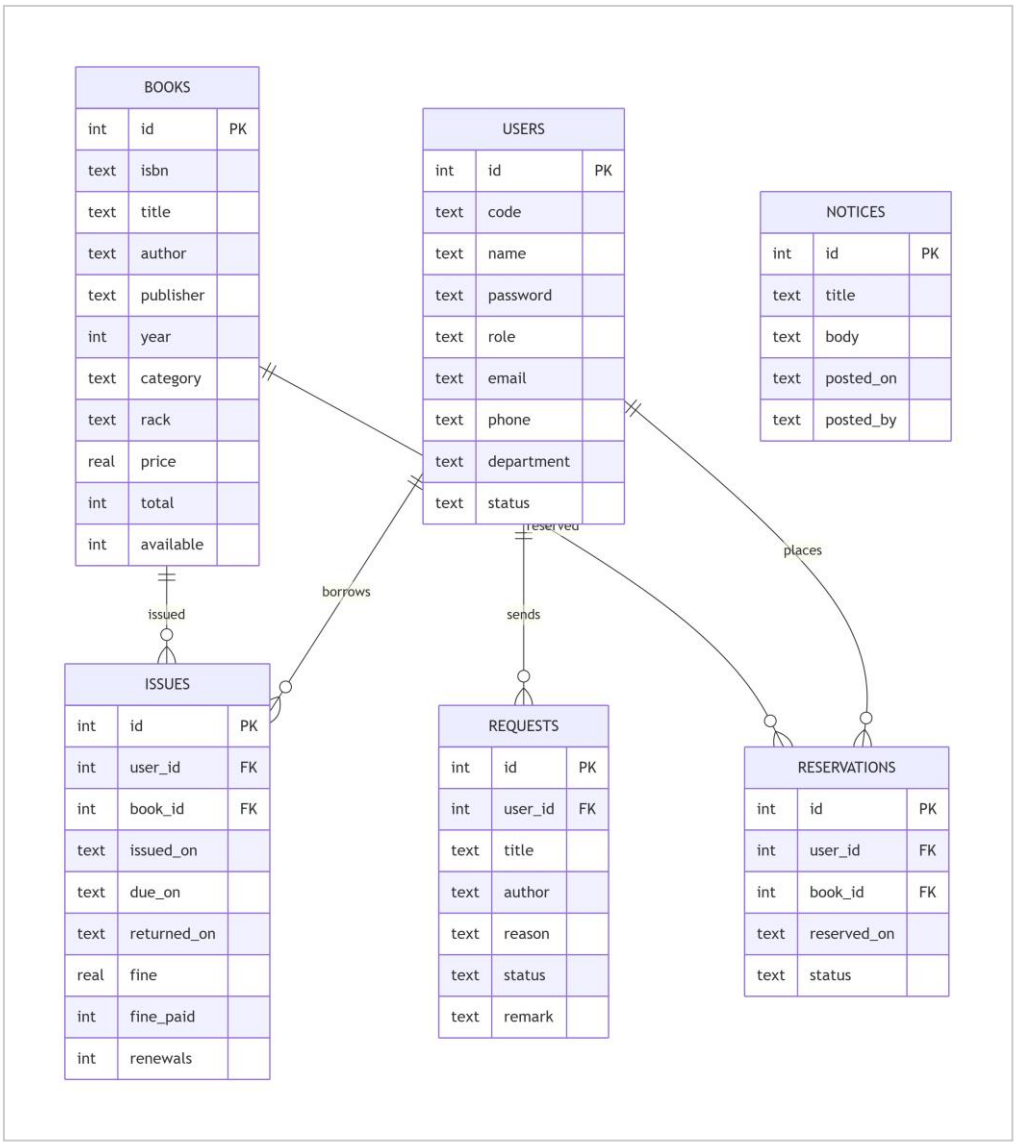


Figure 2: Entity-Relationship Diagram

4.3 Class Diagram

LibraryApp manages screen switching. PortalShell is the base class for all three role portals. Database provides all data functions.

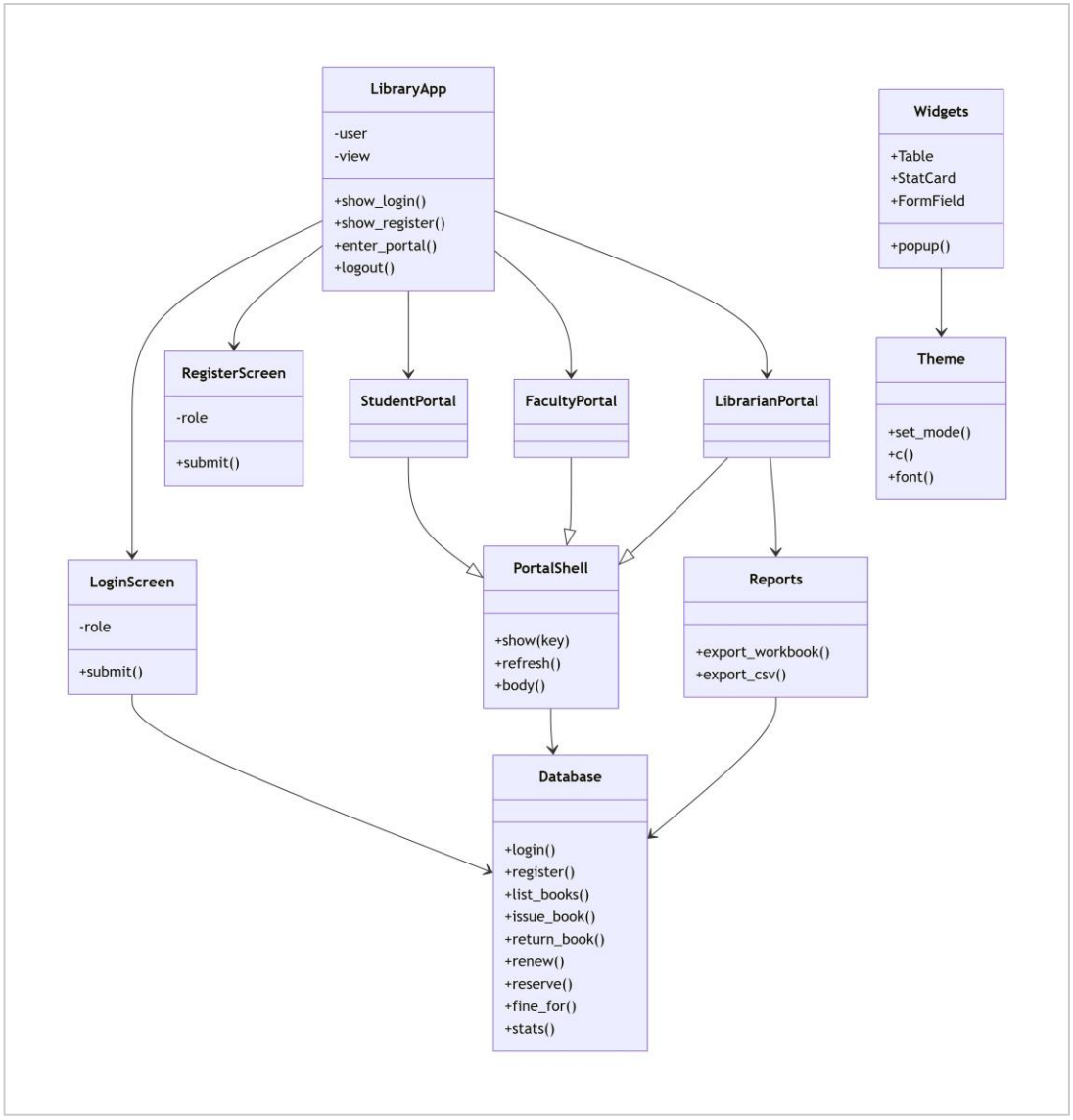


Figure 3: UML Class Diagram

4.4 Use Case Diagram

Three actors — Student, Faculty and Librarian — with 13 use cases.

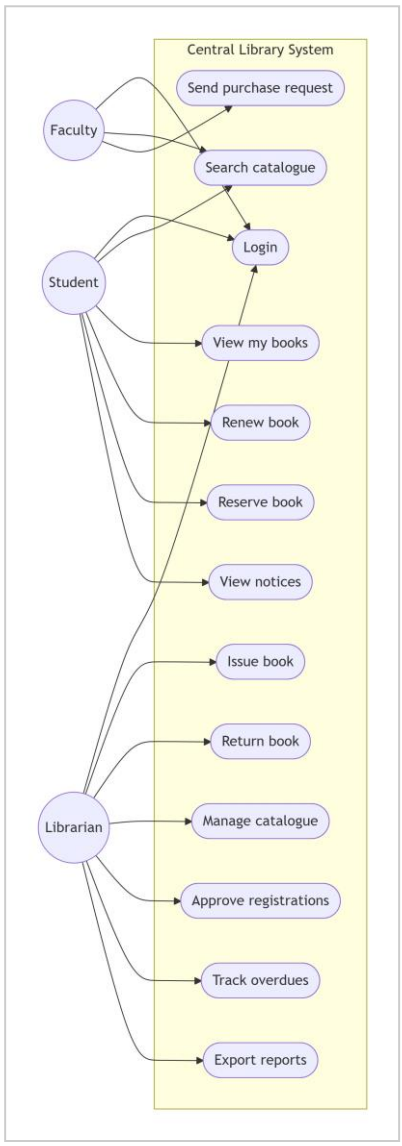


Figure 4: Use Case Diagram

4.5 DFD Level 0

Context diagram showing the system as a single process exchanging data with three actors.

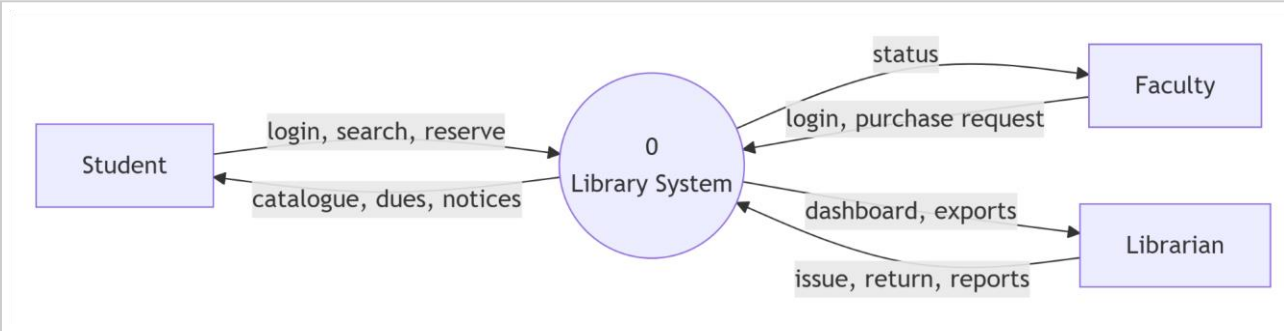


Figure 5: Data Flow Diagram — Level 0

4.6 DFD Level 1

Six processes: Authentication, Catalogue, Circulation, Reservations, Notices & Requests, Reports.

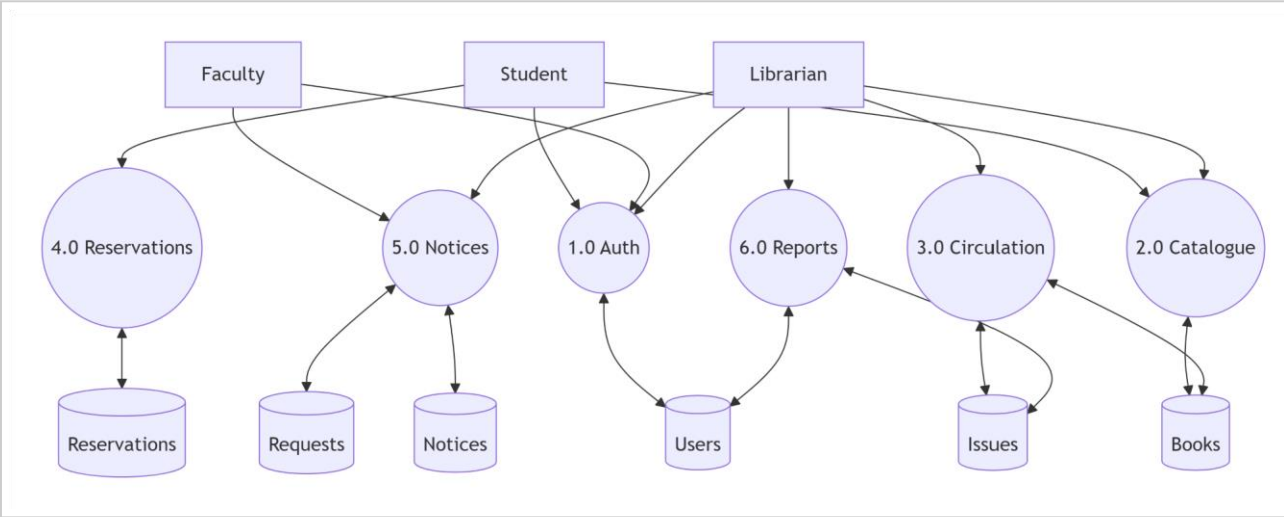


Figure 6: Data Flow Diagram — Level 1

4.7 DFD Level 2 (Circulation)

Sub-processes — lookup, policy check, issue, return and renew.

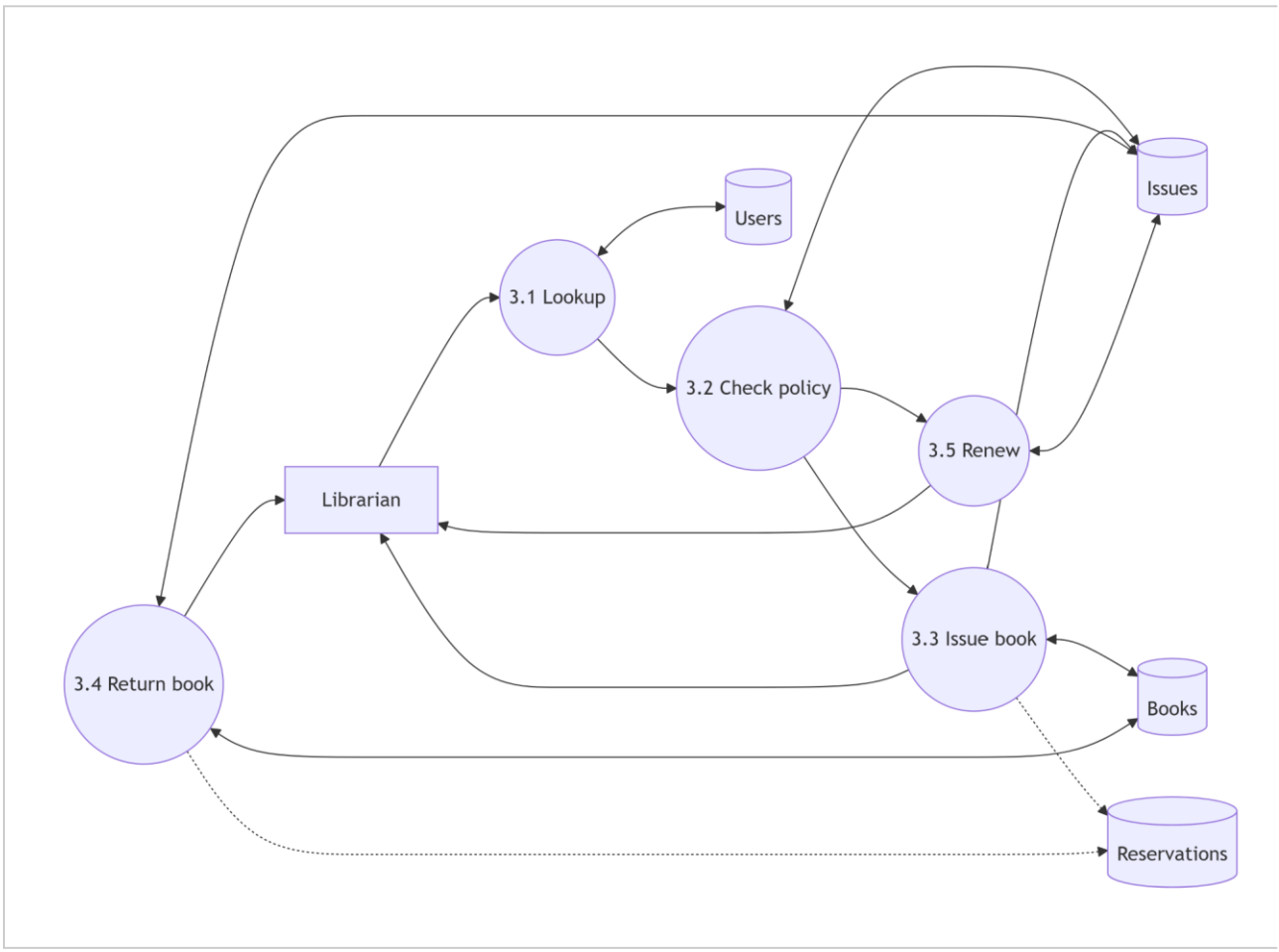


Figure 7: Data Flow Diagram — Level 2

4.8 Sequence Diagrams

4.8.1 Login

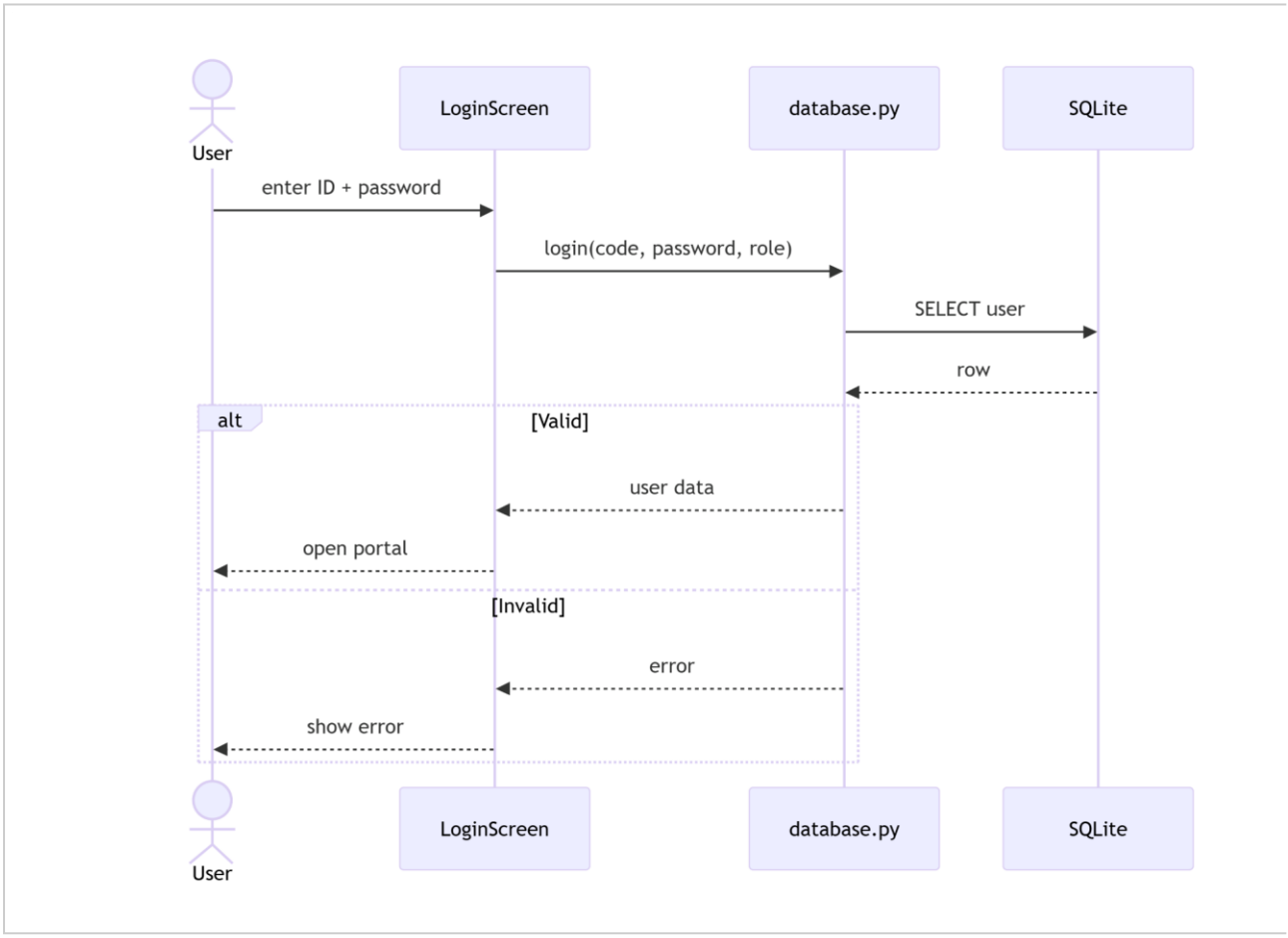


Figure 8a: Sequence Diagram — Login

4.8.2 Issue Book

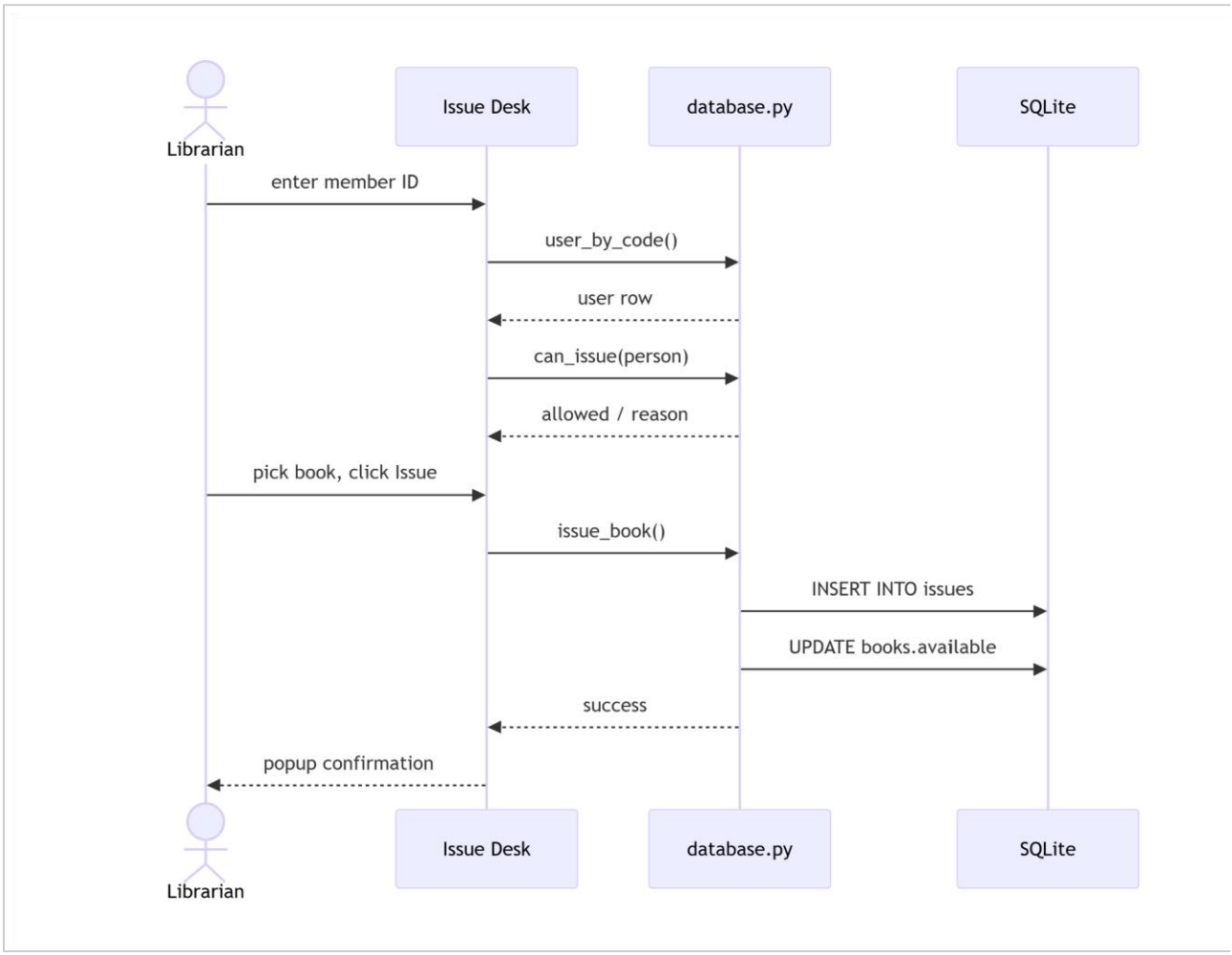


Figure 8b: Sequence Diagram — Issue Book

4.8.3 Return Book

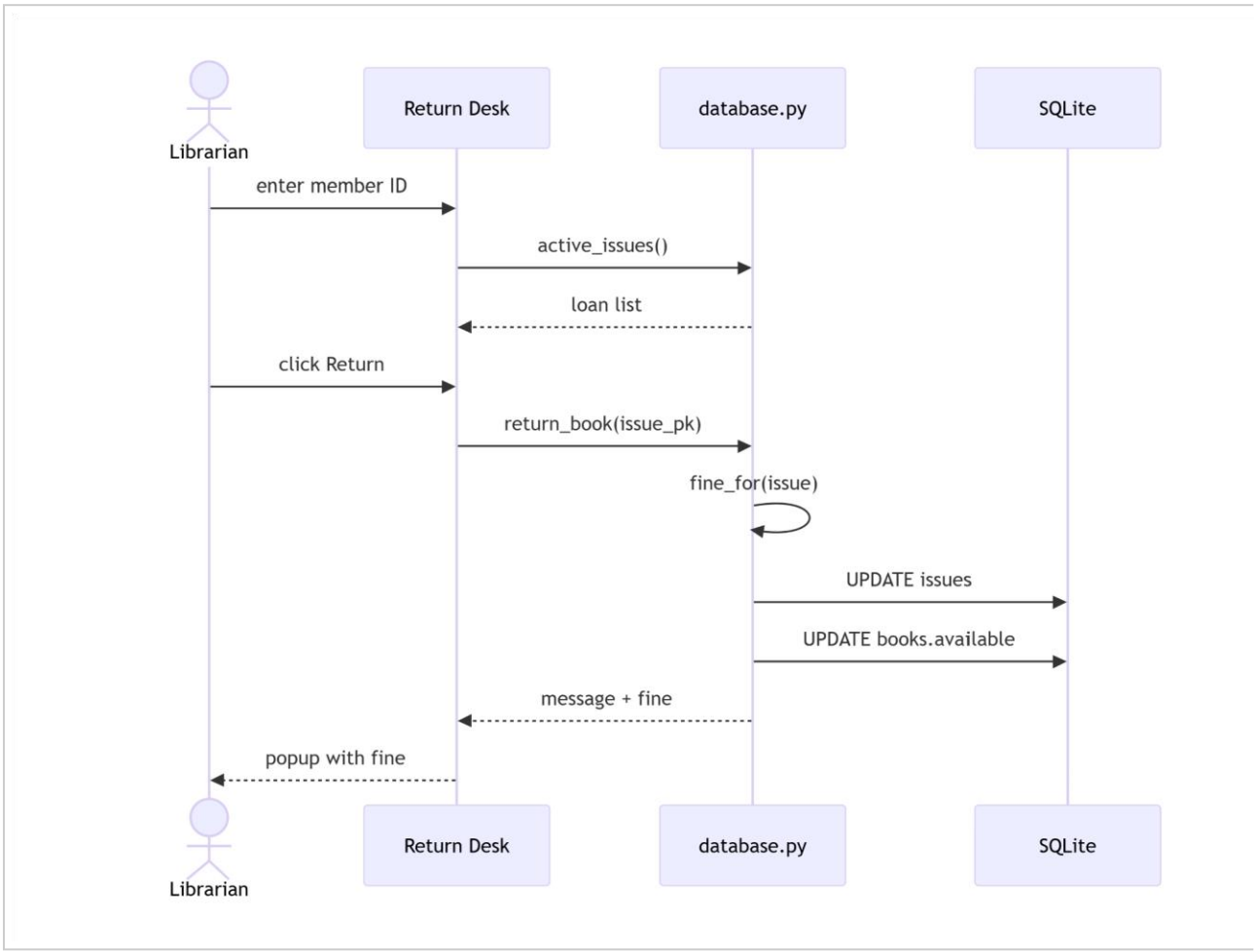


Figure 8c: Sequence Diagram — Return Book

4.9 Activity Diagram

Complete issue → return lifecycle with all decision points.

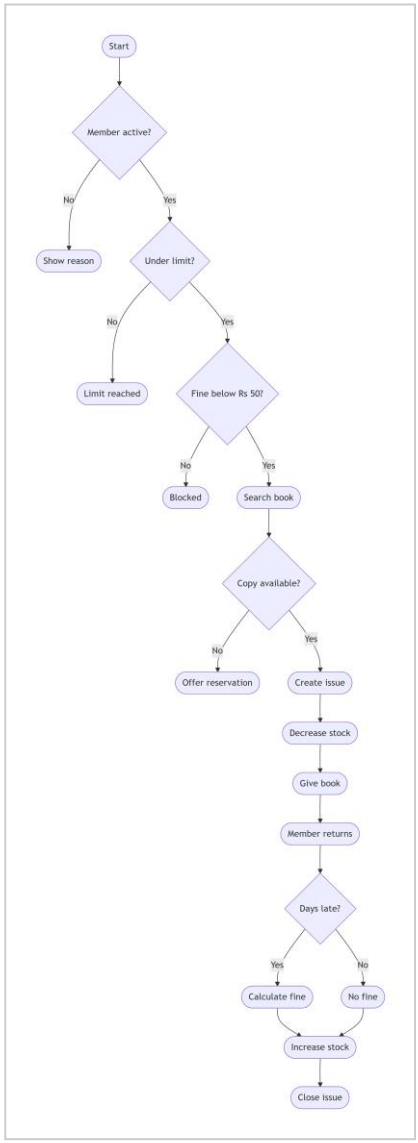


Figure 9: Activity Diagram

4.10 Component Diagram

Shows how the files connect — main.py → screens → database.py.

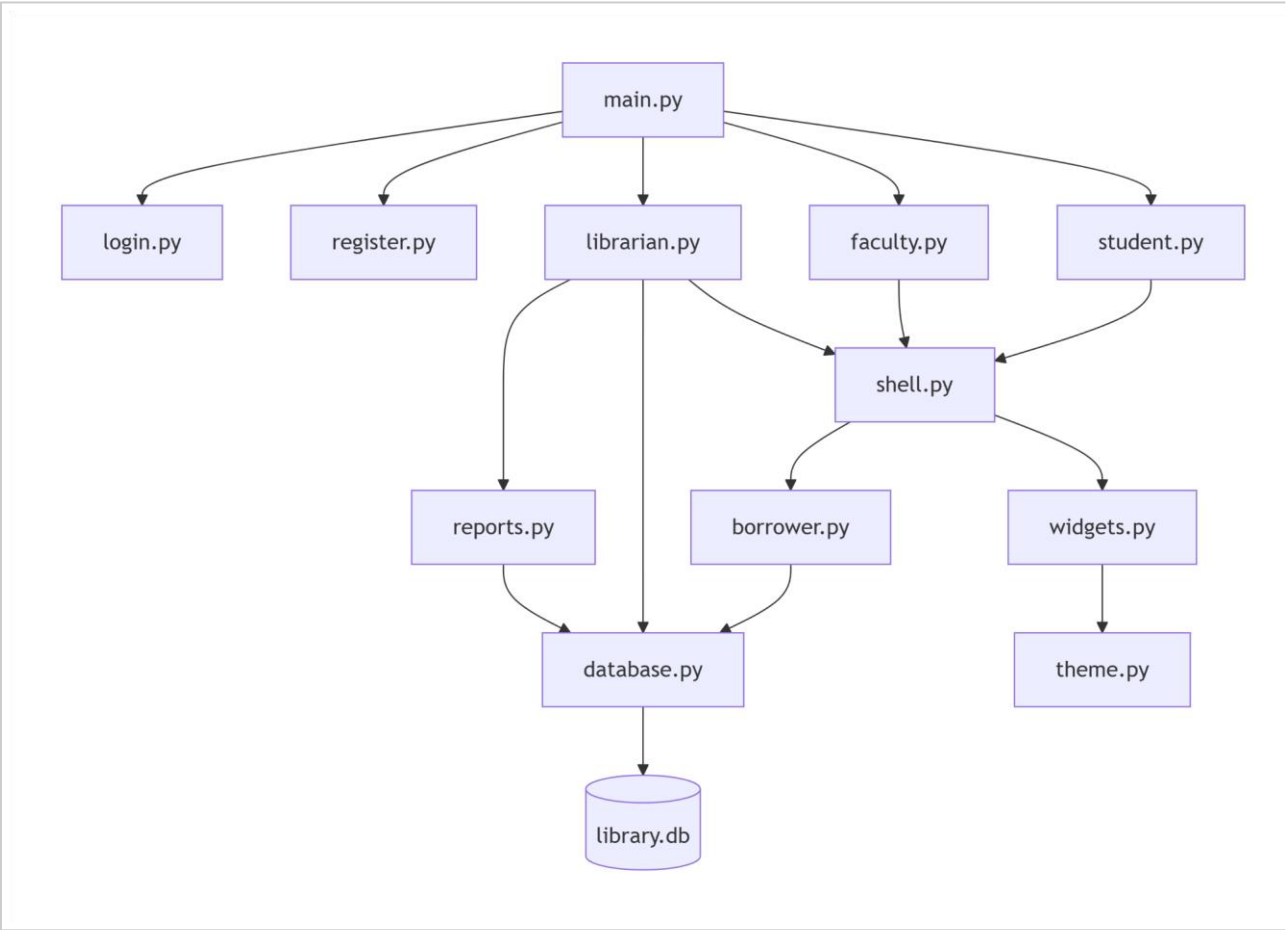


Figure 10: Component Diagram

4.11 Deployment Diagram

Single-tier desktop application. Everything runs on the user's machine.

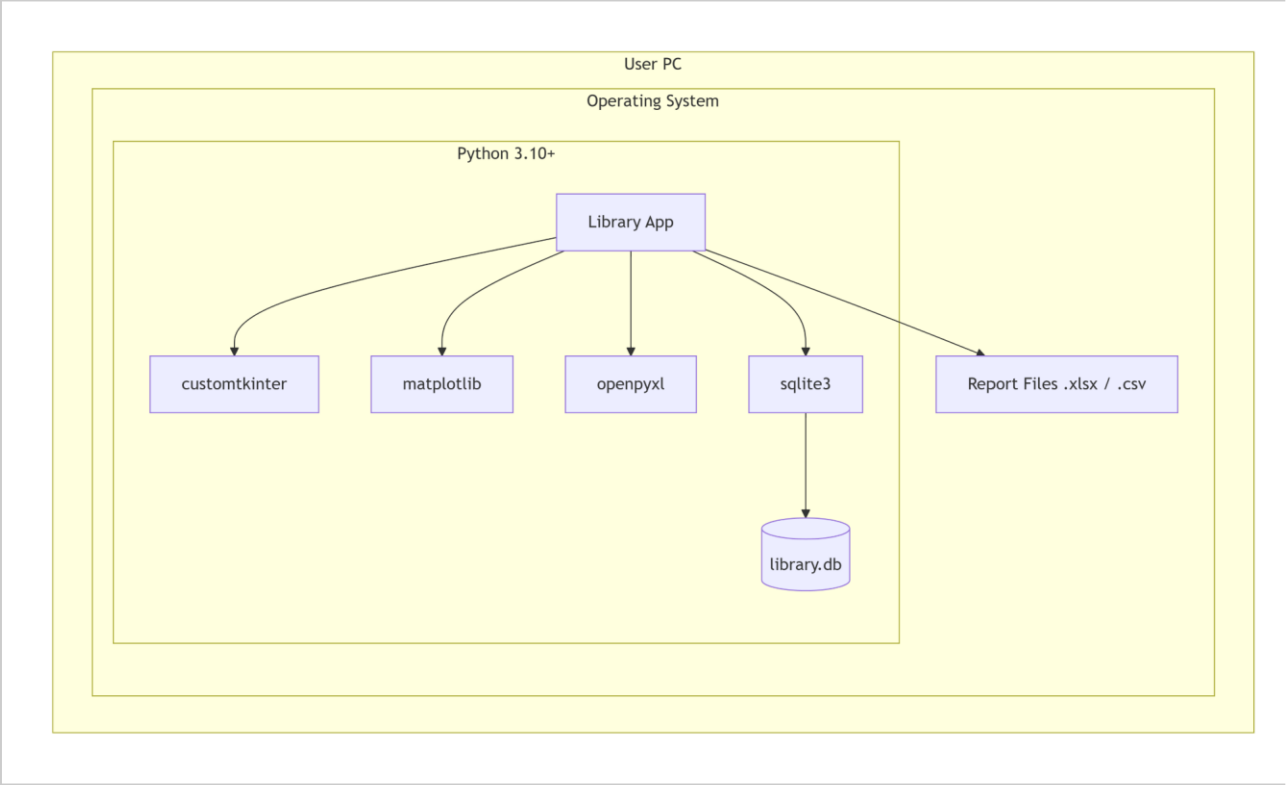


Figure 11: Deployment Diagram

4.12 State Diagrams

4.12.1 User Account

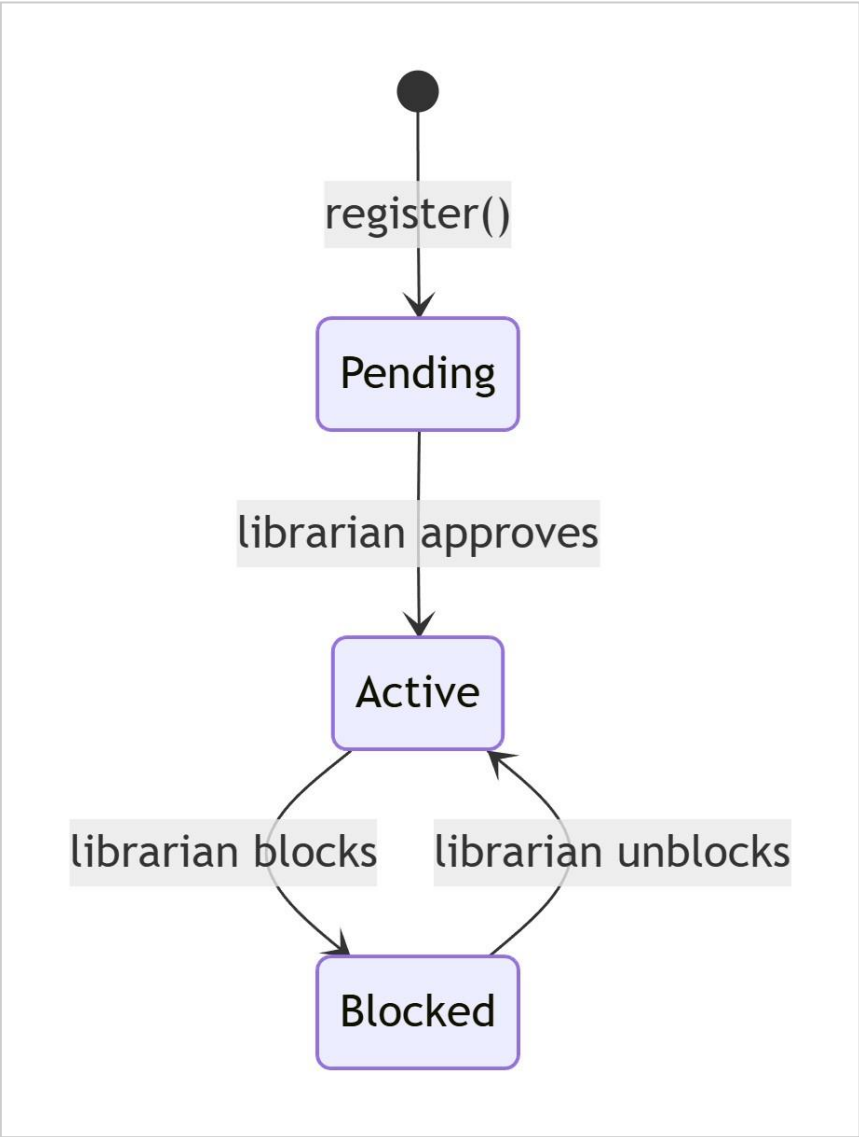


Figure 12a: State Diagram — User Account

4.12.2 Loan

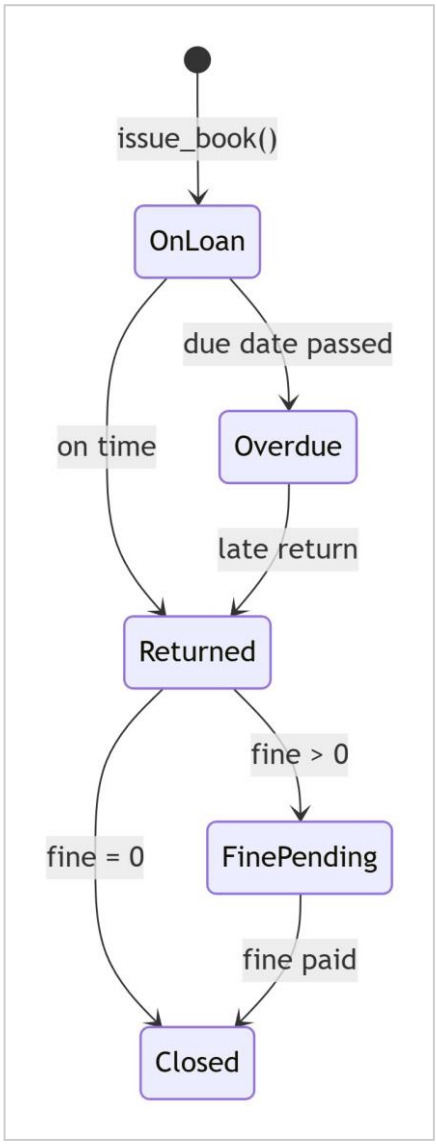


Figure 12b: State Diagram — Book Loan

4.12.3 Reservation

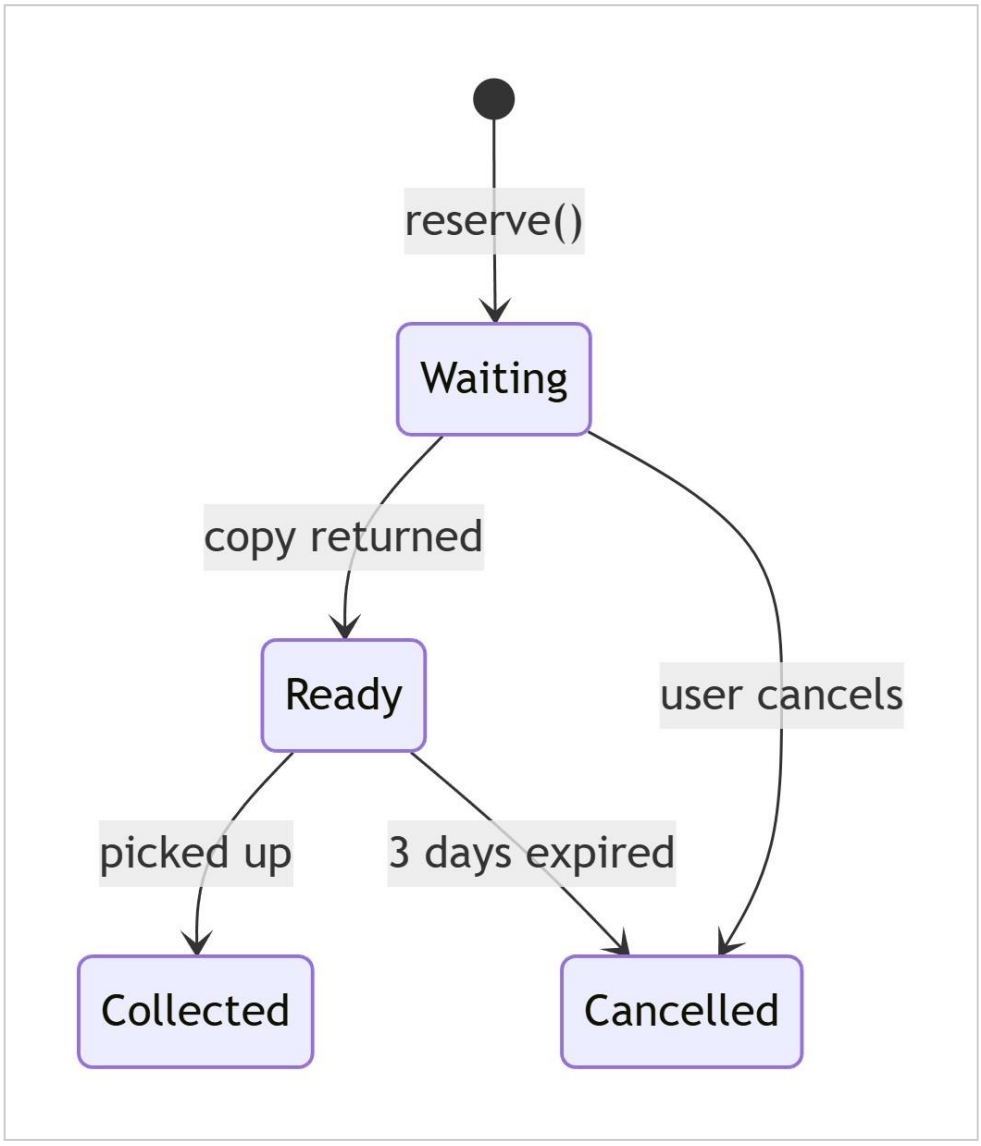


Figure 12c: State Diagram — Reservation

4.13 Package Diagram

Shows the project folder structure.

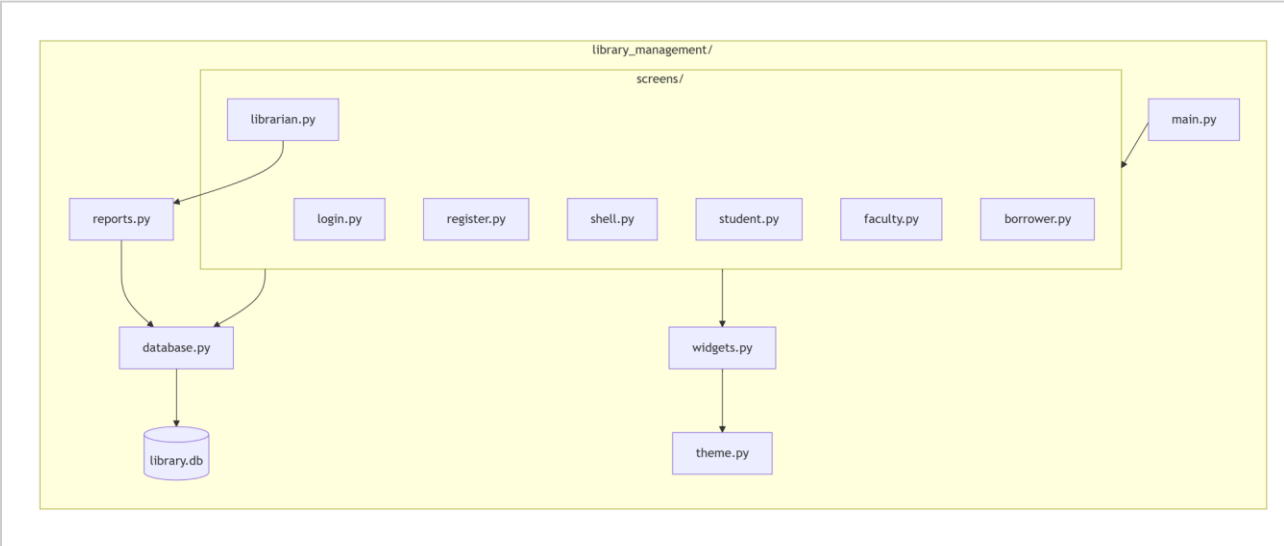


Figure 13: Package Diagram

5. DATABASE DESIGN

Table	Purpose	Key Columns
users	Students, faculty, librarians	id, code, name, role, status
books	Catalogue of all titles	id, isbn, title, author, total, available
issues	Issue/return records with fine	id, user_id, book_id, issued_on, due_on, fine
reservations	Queue of waiting members	id, user_id, book_id, status
notices	Library announcements	id, title, body, posted_on
requests	Faculty purchase requests	id, user_id, title, status

Borrowing Policy (Constants)

```
POLICY = {
    "student":   {"limit": 2, "days": 14},
    "faculty":   {"limit": 5, "days": 60},
"librarian": {"limit": 0, "days": 0},
}
FINE_PER_DAY      = 2     # Rs
FINE_BLOCK_LIMIT  = 50    # Rs
MAX_RENEWALS      = 1
```

6. IMPLEMENTATION

6.1 Module Description

File	Purpose

main.py	Entry point, screen switching
database.py	Schema, seed data, library rules
reports.py	Excel and CSV export
theme.py	Dark and light colour palettes
widgets.py	Reusable UI components
screens/login.py	Login screen
screens/register.py	Registration screen
screens/shell.py	Common sidebar + topbar
screens/student.py	Student portal
screens/faculty.py	Faculty portal
screens/librarian.py	Librarian portal
screens/borrower.py	Shared borrower pages

6.2 Screenshots

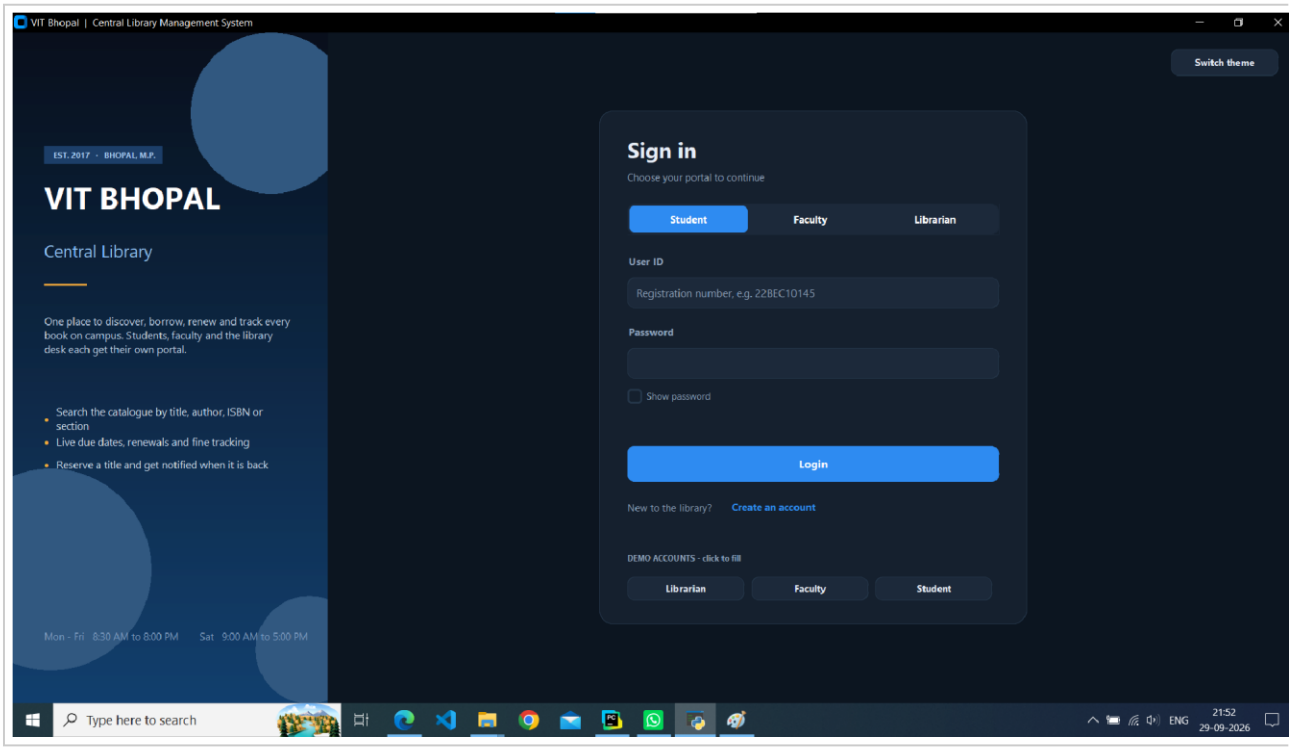


Figure 14: Login Screen

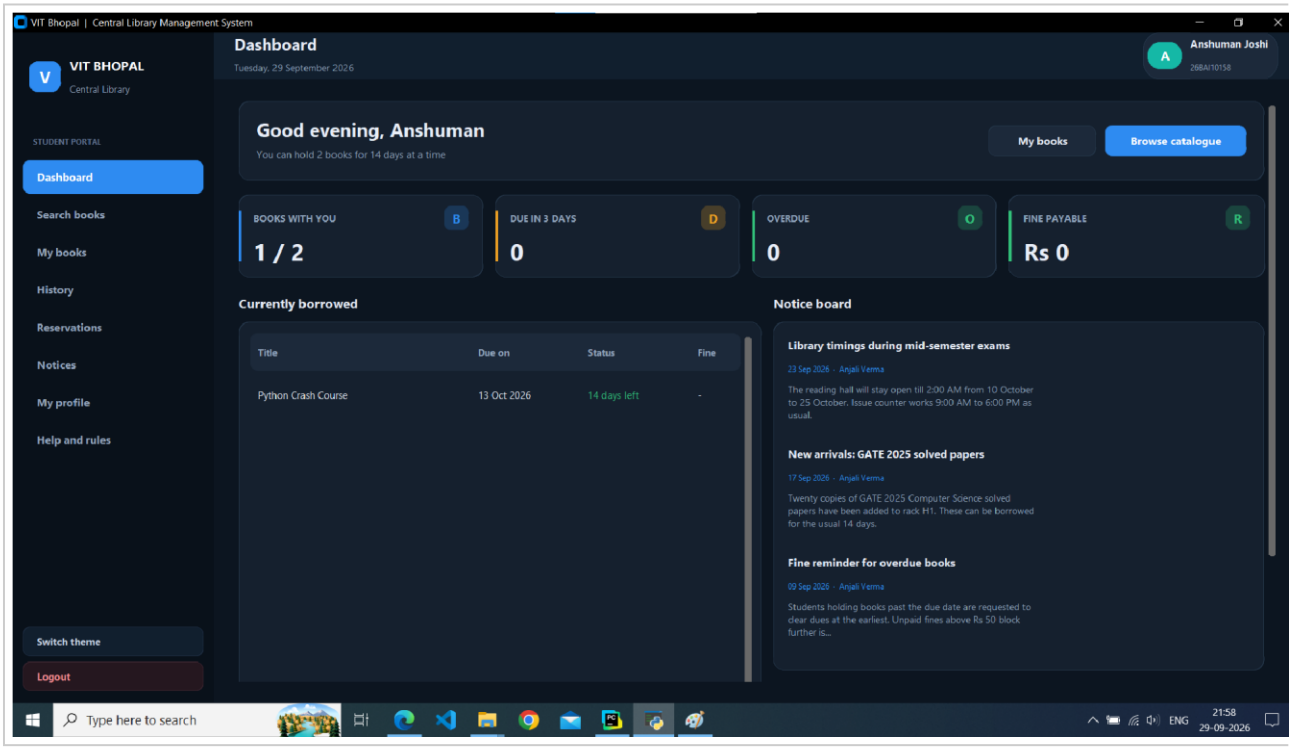


Figure 15: Student Portal — Dashboard

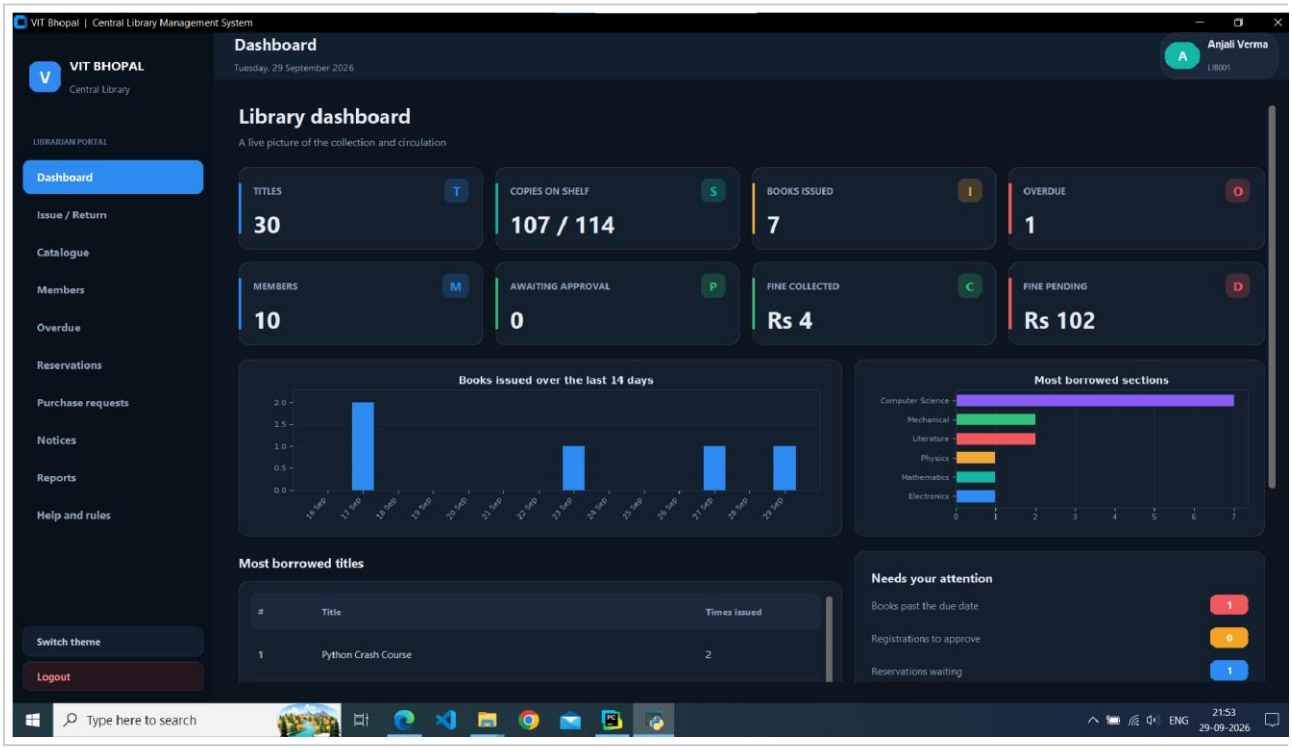


Figure 16: Librarian Portal — Dashboard

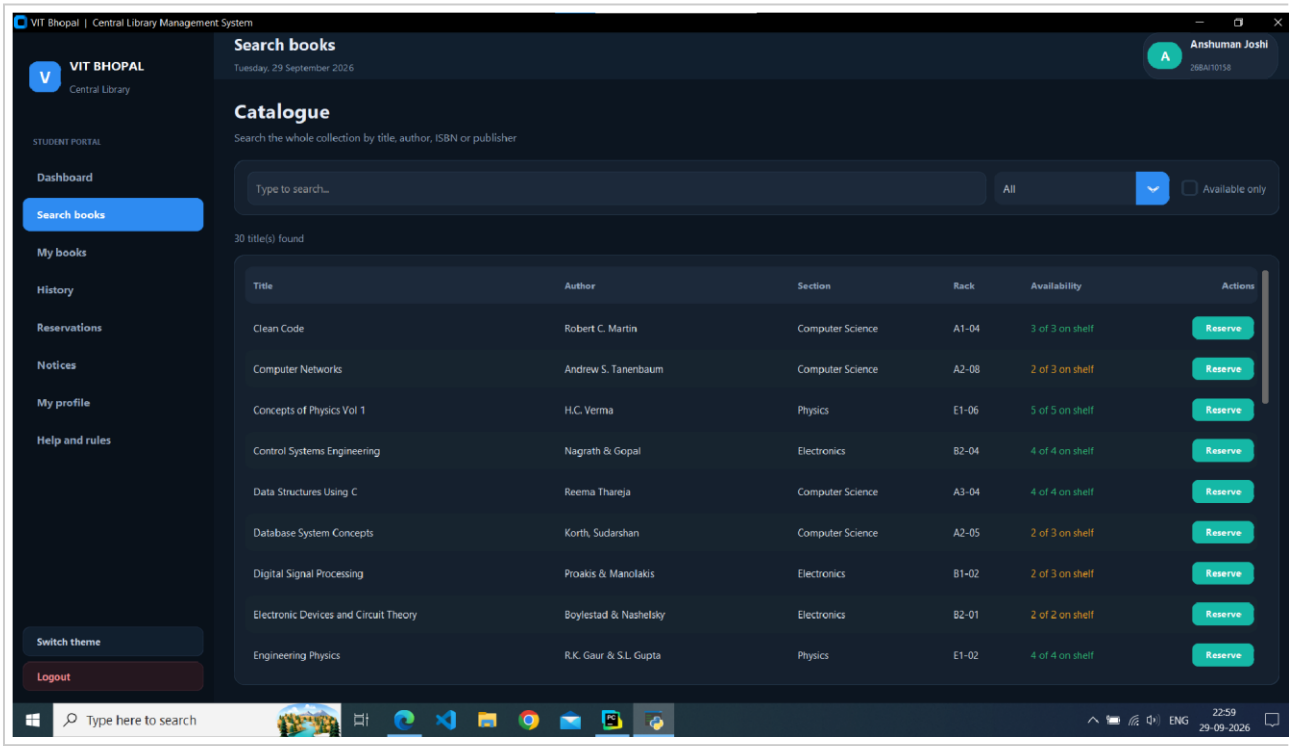


Figure 17: Catalogue Search

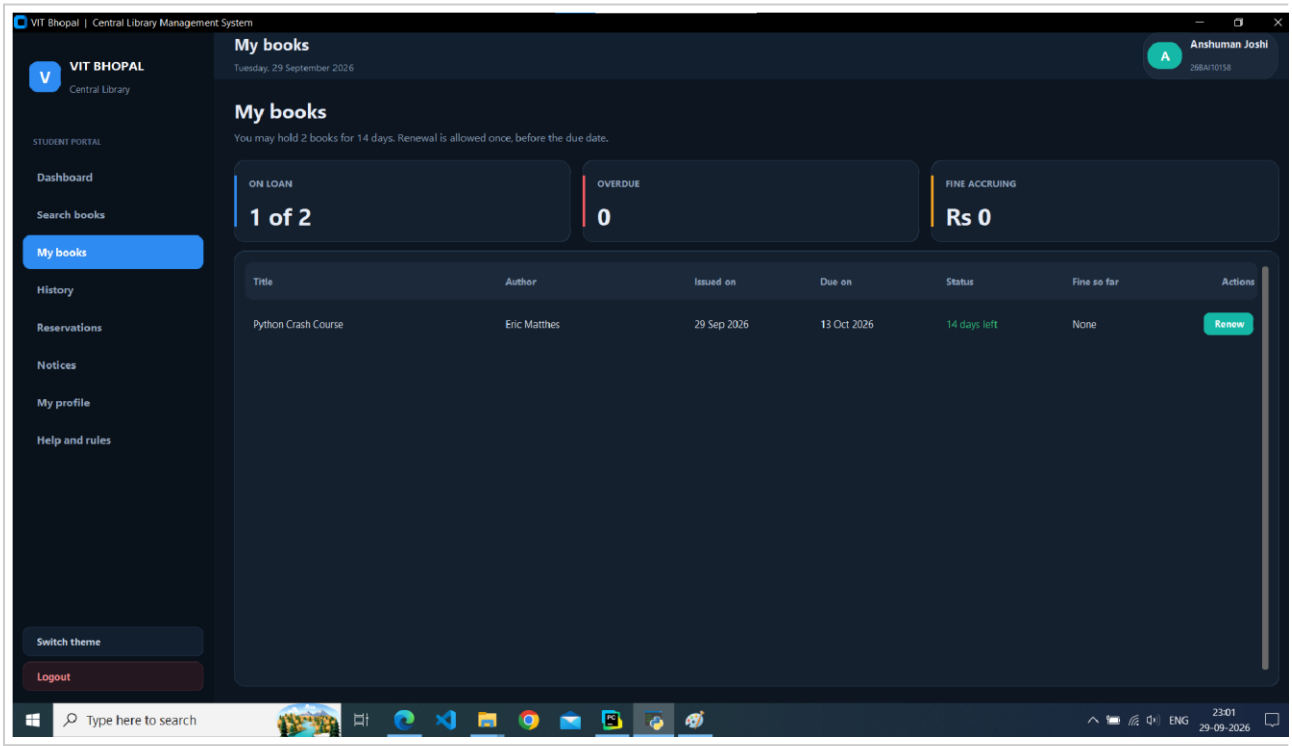


Figure 18: My Books (Loans)

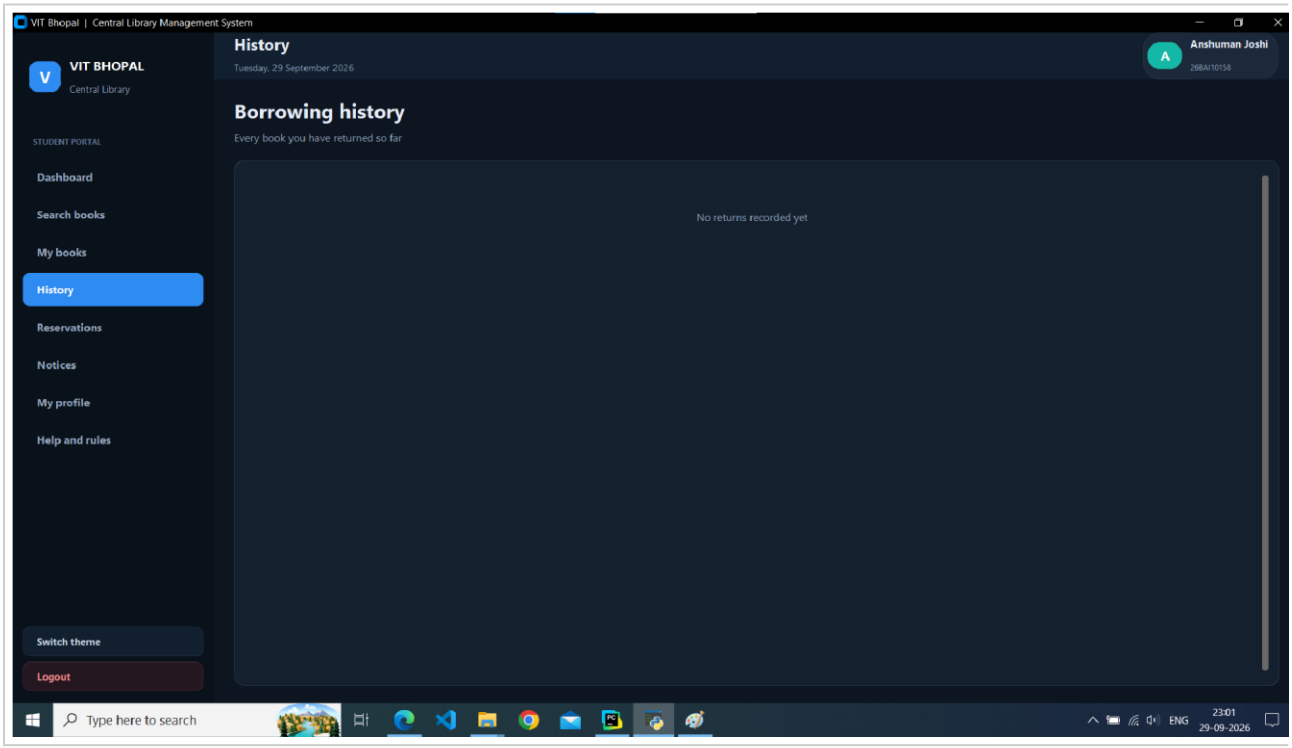


Figure 19: Borrowing History

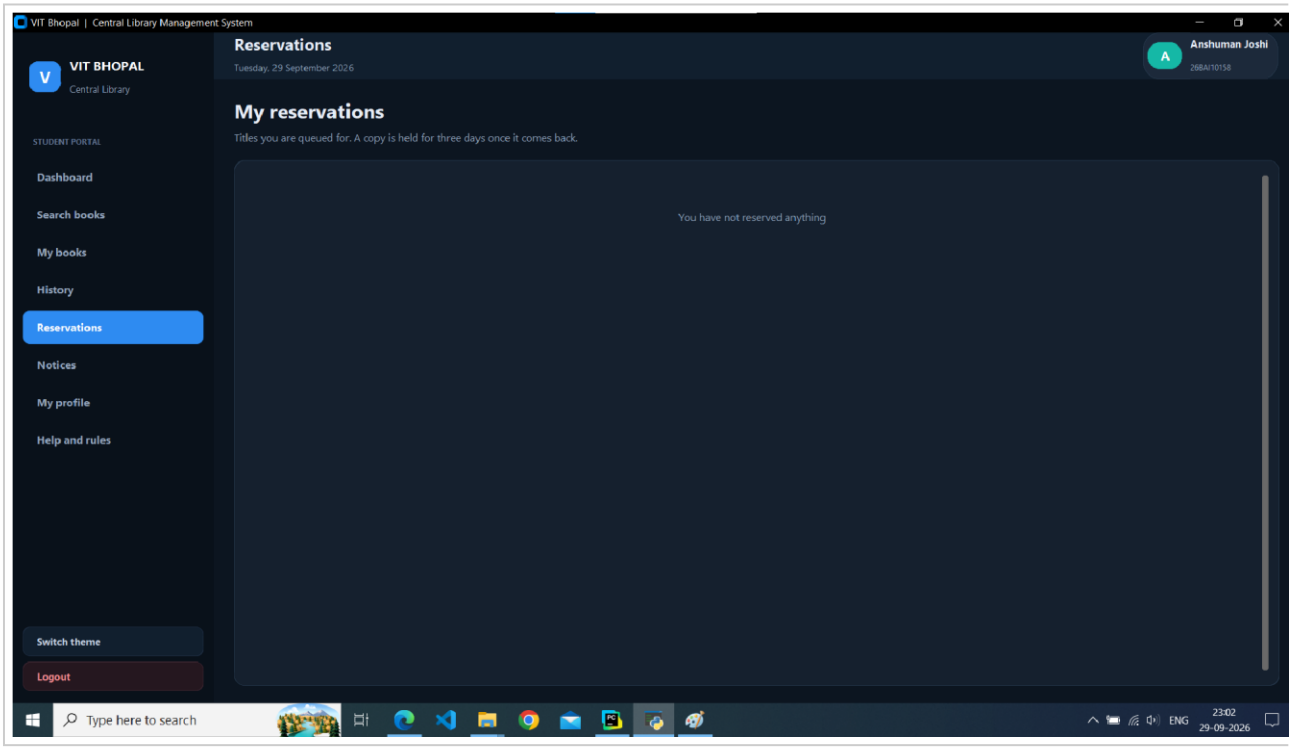


Figure 20: My Reservations

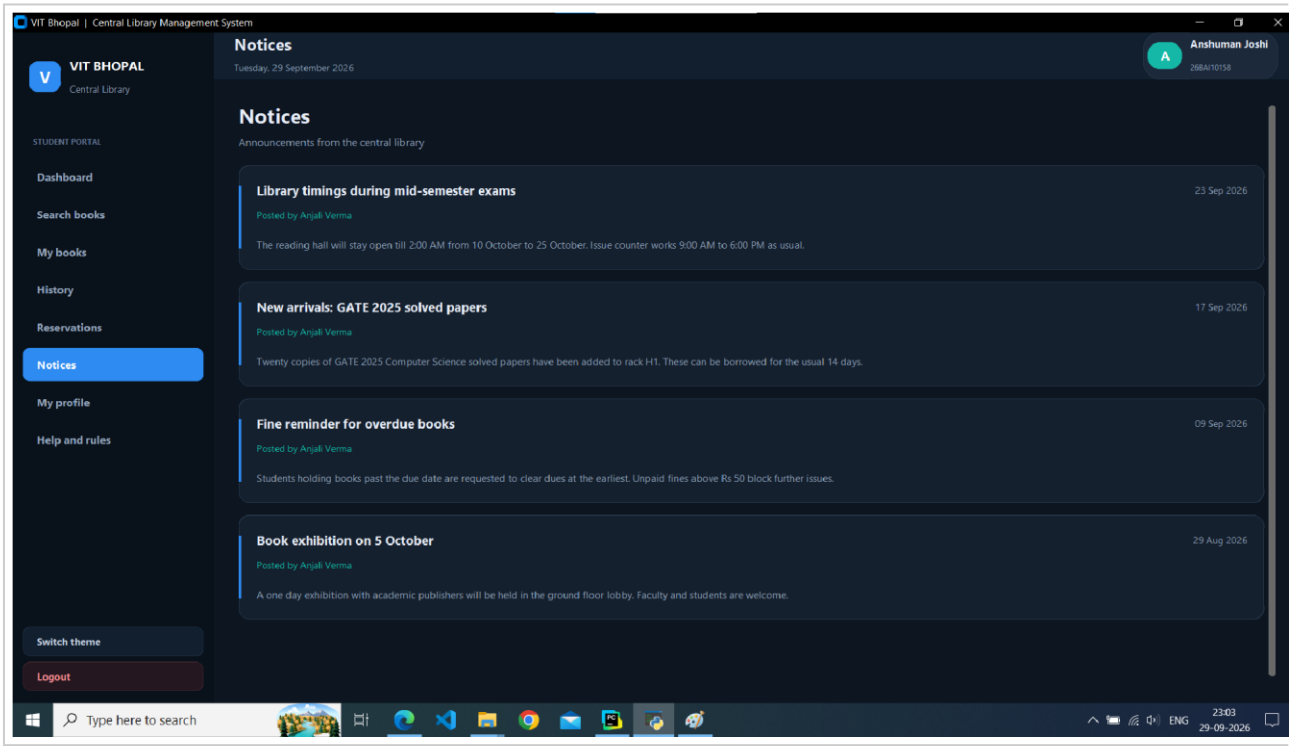


Figure 21: Notice Board

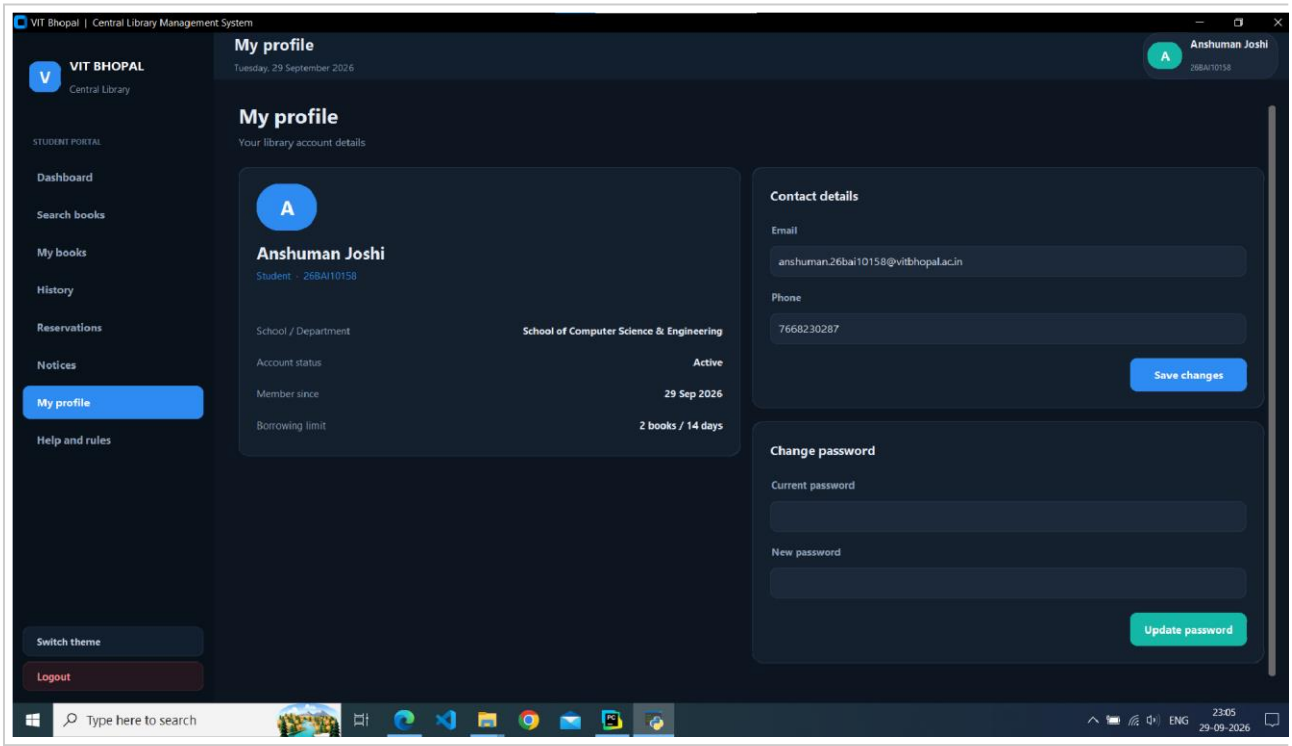


Figure 22: My Profile

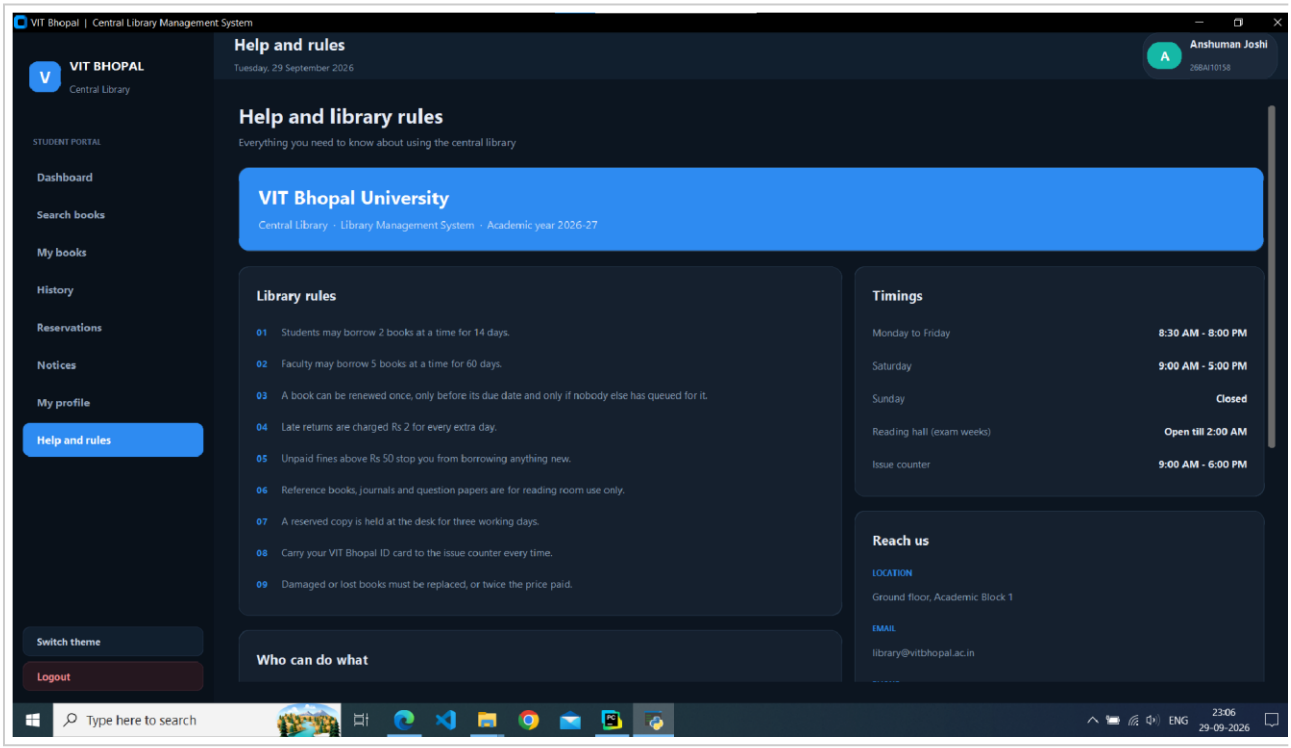


Figure 23: Help & Rules

6.3 Key Code — Fine Calculation

```
FINE_PER_DAY = 2

def fine_for(issue, on=None):
    due = date.fromisoformat(issue["due_on"])     end =
(date.fromisoformat(issue["returned_on"])            if
issue.get("returned_on") else (on or date.today()))     late =
(end - due).days     return late * FINE_PER_DAY if late > 0
else 0
```

7. TESTING

TC#	Test Case	Expected Result	Status
TC01	Student login valid	Portal opens	Pass
TC02	Login wrong password	Error shown	Pass
TC03	Borrow 3rd book (limit 2)	Rejected	Pass
TC04	Issue with Rs 60 fine	Blocked	Pass
TC05	Issue available copy	Success, stock -1	Pass
TC06	Return 5 days late	Fine = Rs 10	Pass
TC07	Renew before due date	+14 days	Pass
TC08	Renew twice	Second rejected	Pass
TC09	Reserve unavailable book	Status = waiting	Pass
TC10	Return with reservation	Next waiter ready	Pass
TC11	New registration	Status = pending	Pass
TC12	Approve pending account	Status = active	Pass
TC13	Export xlsx report	6-sheet file downloaded	Pass
TC14	Export transactions CSV	CSV downloaded	Pass
TC15	Switch theme	Dark ↔ Light	Pass

Test Accounts

- Librarian: `LIB001` / `admin123`
- Student: `22BEC10145` / `student123`
- Faculty: `EMP1042` / `faculty123`

8. RESULTS AND CONCLUSION

8.1 Achievements

- Fully functional system with three role-based portals
- Automatic fine calculation and borrowing-limit enforcement
- Reservation queue with automatic promotion
- Excel and CSV report generation
- Modern dark/light UI
- Clean three-layer architecture

8.2 Limitations

- Local-only database
- No email/SMS notifications
- No barcode/RFID support

8.3 Future Scope

- Email/SMS due-date reminders
- Online payment gateway for fines
- Barcode/RFID support
- Web and mobile versions
- Cloud backup
- Multi-branch library support

9. REFERENCES

1. Python Software Foundation — *Python Documentation*, https://python.org
2. Tom Schimansky — *CustomTkinter Docs*, https://customtkinter.tomschimansky.com
3. SQLite Consortium — *SQLite Documentation*, https://sqlite.org/docs.html
4. Matplotlib — *Documentation*, https://matplotlib.org
5. openpyxl — *Documentation*, https://openpyxl.readthedocs.io
6. VIT Bhopal — *Central Library Handbook*, 2024
7. Ian Sommerville — *Software Engineering*, 10th Ed., Pearson, 2016

— END OF REPORT —